First, although the bars built by Weber, by Braginsky, and by others had achieved far better sensitivities than anyone had dreamed possible in the 1950s, they were still only able to detect with confidence waves of strength 10^{-17} or larger. This was 10,000 times too poor for success, if I and others had correctly estimated the strengths of the waves arriving at Earth. This by itself was not serious, since the march of technology has often produced 10,000-fold improvements in instruments over times of twenty years or less. [One example was the angular resolution of the best radio telescopes, which improved from tens of degrees in the mid-1940s to a few arc seconds in the mid-1960s (Chapter 9). Another was the sensitivity of astronomical X-ray detectors, which improved by a factor of 10^{10} between 1958 and 1978, that is, at an average rate of 10,000 every eight years (Chapter 8).] However, the rate of improvement of the bars was so slow, and projections of the future technology and techniques were so modest, that there seemed no reasonable way for a 10,000-fold improvement to be made in the foreseeable future. Success, thus, would likely hinge on the waves being stronger than the 10^{-21} estimates—a real possibility, but not one that anybody was happy to rely on.

Second, even if the bars did succeed in detecting gravitational waves, they would have enormous difficulty in decoding the waves' symphonic signals, and in fact would probably fail. The reason was simple: Just as a tuning fork or wine glass responds sympathetically only to a sound whose frequency is close to its natural frequency, so a bar would respond only to gravitational waves whose frequency is near the bar's natural frequency; in technical language, the bar detector has a *narrow bandwidth* (the bandwidth being the band of frequencies to which it responds). But the waves' symphonic information should typically be encoded in a very wide band of frequencies. To extract the waves' information, then, would require a "xylophone" of many bars, each covering a different, tiny portion of the signal's frequencies. How many bars in the xylophone? For the types of bars then being planned and constructed, several thousand—far too many to be practical. In principle it would be possible to widen the bars' bandwidths and thereby manage with, say, a dozen bars, but to do so would require major technical advances beyond those for reaching a sensitivity of 10^{-21}.

Although I did not say much in public in the 1980s about my pessimistic outlook, in private I regarded it as tragic because of the great